Structural Self-Energy Expansion (SSEE-V3.6)

A zero-free-parameter dark energy model from φ and π

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Seeking arXiv endorsement in astro-ph.CO (or gr-qc)

Core idea in one sentence

All cosmological observables — w_0, w_a, H_0, Ω_m, α_K, m_φ, n_s, β_c — are derived *algebraically* from the golden ratio $\varphi = (1+\sqrt{5})/2$ and π with **zero free parameters**.

Key algebraic predictions

Quantity	SSEE algebraic value	Observed / constraint	Agreement
w_0	$-T_r/M_v = -0.8400$	DESI DR2: -0.827 ± 0.062	0.21σ
w_a	$-P_{sc}/K_v = -0.6699$	DESI DR2: $-0.75^{+0.29}_{-0.25}$	0.31σ
H_0	$3(\varphi+\pi)^2 = 67.96$ km/s/Mpc	Planck 2018: 67.36 ± 0.54	1.11σ
r_d	CAMB+MIRA: 147.2 Mpc	Planck 2018: 147.09 ± 0.26 Mpc	0.25σ
n_s	$1 - \varphi^{-7} = 0.9656$	Planck PR4: 0.9649 ± 0.0042	0.17σ
$\alpha_K(0)$	$3\Omega_{\rm DE}(1+w_0) = 0.4033$	CLASS cross-check: 0.4033	0.005%
m_φ	$\Sigma m_\nu \cdot H_0^{\rm alg} = 5.71$ eV	KATRIN/PTOLEMY (2027–30)	Prediction
$k_{\rm fs}$	$0.493\ h$/Mpc	DESI Y3/Euclid (2026–28)	Prediction

Seven-paper series (all reproducible)

1. **Framework + EFT**: Algebraic derivation of all constants; Bellini–Sawicki α-functions; α-attractor $\alpha = \varphi^4/3$; $\beta_c = -\text{AURA} = -3.998$ (plateau test). [22 pp]
2. **Bayesian MCMC**: 100 walkers × 25 000 steps vs. DESI DR2 + Planck + 7 clusters; $\Delta\text{BIC} = -5.55$ (SSEE favoured, $k_{\rm SSEE} = 1$ vs. $k_{\Lambda\rm CDM} = 3$, dynamic sector). [21 pp]
3. **CMB confrontation**: CAMB + MIRA two-sector mapping; $\chi^2_r(\text{TT}) = 1.062$, $\chi^2_r(\text{TE}) = 1.053$; CLASS cross-check α_K to 0.005%. [18 pp]
4. **Nine Sovereignties**: All CMB observables (ℓ_1, ℓ_2, ℓ_3, θ_*, A_s, n_s, τ, r) derived from φ, π only. [15 pp]
5. **Israel–Stewart perturbations**: Causal bulk viscosity; $c_s^2 = 0$ exact; MIRA is background-sector effect (not perturbative); $\gamma_{\rm IS} = 0.554 \approx \gamma_{\Lambda\rm CDM}$. [24 pp]
6. **φ-DM two-sector**: $m_\varphi = 5.71$ eV algebraic; $\Omega_{\varphi\rm DM} = (\text{MIRA} - 1)\Omega_{m,\rm dyn}$; $f\sigma_8$ tension $3.66\sigma \to 0.50\sigma$ resolved. [16 pp]
7. **Canonical EFT**: $\beta_c = -\text{AURA}$ (numerical -3.990, gap 0.2% attributed to simplified background); $\alpha_T = \alpha_M = \alpha_B = 0$ exact (GW170817 consistent); $\alpha_K = 0.4033$ algebraic; λ, V_0, M, g^2 all blocked. [14 pp]

Falsifiability (explicit)

- $m_\varphi = 5.71$ eV — KATRIN (2027) or PTOLEMY (2030) will confirm or refute.
- $k_{\rm fs} = 0.493\ h$/Mpc — DESI Y3 + Euclid lensing will confirm or refute.
- $\alpha_K = 0.403$ — Euclid spectroscopic growth rate will confirm or refute.
- Any deviation of w_0 from -0.840 by $> 3\sigma$ refutes the framework.

What I am asking for

An arXiv endorsement in `astro-ph.CO` (or `gr-qc`) to submit Papers 1–3 as the first tranche. All code, data, and LaTeX sources are public at github.com/mikealmeida1721/SSEE under Apache-2.0.

I am happy to answer any technical questions or share the PDFs directly. Thank you for considering this request.

Code: Python (emcee, CAMB, CLASS), all reproducible in <30 min. **Data:** DESI DR2 (public), Planck PR4 (public), KiDS-1000 (public).